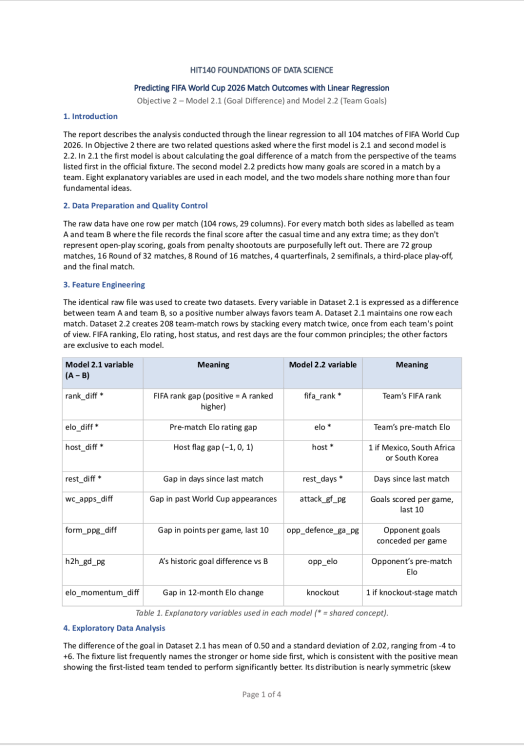
Digital Receipt

This receipt acknowledges that Turnitin received your paper. Below you will find the receipt information regarding your submission.

The first page of your submissions is displayed below.

Submission author:	MOU BISWAS
Assignment title:	Turnitin checker_HIT140_Assessment 3
Submission title:	Foundation Of Data Science Assesment3.docx
File name:	Foundation_Of_Data_Science_Assesment3.docx
File size:	228.18K
Page count:	4
Word count:	1,620
Character count:	8,435
Submission date:	15-Oct-2026 10:47AM (UTC+0930)
Submission ID:	3060141989



HIT140 FOUNDATIONS OF DATA SCIENCE

Predicting FIFA World Cup 2026 Match Outcomes with Linear Regression

Objective 2 – Model 2.1 (Goal Difference) and Model 2.2 (Team Goals)

1. Introduction

The report describes the analysis conducted through the linear regression to all 104 matches of FIFA World Cup 2026. In Objective 2 there are two related questions asked where the first model is 2.1 and second model is 2.2. In 2.1 the first model is about calculating the goal difference of a match from the perspective of the teams listed first in the official fixture. The second model 2.2 predicts how many goals are scored in a match by a team. Eight explanatory variables are used in each model, and the two models share nothing more than four fundamental ideas.

2. Data Preparation and Quality Control

The raw data have one row per match (104 rows, 29 columns). For every match both sides as labelled as team A and team B where the file records the final score after the casual time and any extra time; as they don't represent open-play scoring, goals from penalty shootouts are purposefully left out. There are 72 group matches, 16 Round of 32 matches, 8 Round of 16 matches, 4 quarterfinals, 2 semifinals, a third-place play-off, and the final match.

3. Feature Engineering

The identical raw file was used to create two datasets. Every variable in Dataset 2.1 is expressed as a difference between team A and team B, so a positive number always favors team A. Dataset 2.1 maintains one row each match. Dataset 2.2 creates 208 team-match rows by stacking every match twice, once from each team's point of view. FIFA ranking, Elo rating, host status, and rest days are the four common principles; the other factors are exclusive to each model.

Model 2.1 variable (A – B)	Meaning	Model 2.2 variable	Meaning
rank_diff *	FIFA rank gap (positive = A ranked higher)	fifa_rank *	Team's FIFA rank
elo_diff *	Pre-match Elo rating gap	elo *	Team's pre-match Elo
host_diff *	Host flag gap (−1, 0, 1)	host *	1 if Mexico, South Africa or South Korea
rest_diff *	Gap in days since last match	rest_days *	Days since last match
wc_apps_diff	Gap in past World Cup appearances	attack_gf_pg	Goals scored per game, last 10
form_ppg_diff	Gap in points per game, last 10	opp_defence_ga_pg	Opponent goals conceded per game
h2h_gd_pg	A's historic goal difference vs B	opp_elo	Opponent's pre-match Elo
elo_momentum_diff	Gap in 12-month Elo change	knockout	1 if knockout-stage match

Table 1. Explanatory variables used in each model (= shared concept).*

4. Exploratory Data Analysis

The difference of the goal in Dataset 2.1 has mean of 0.50 and a standard deviation of 2.02, ranging from -4 to +6. The fixture list frequently names the stronger or home side first, which is consistent with the positive mean showing the first-listed team tended to perform significantly better. Its distribution is nearly symmetric (skew

Page 1 of 4

Copyright 2026 Turnitin. All rights reserved.